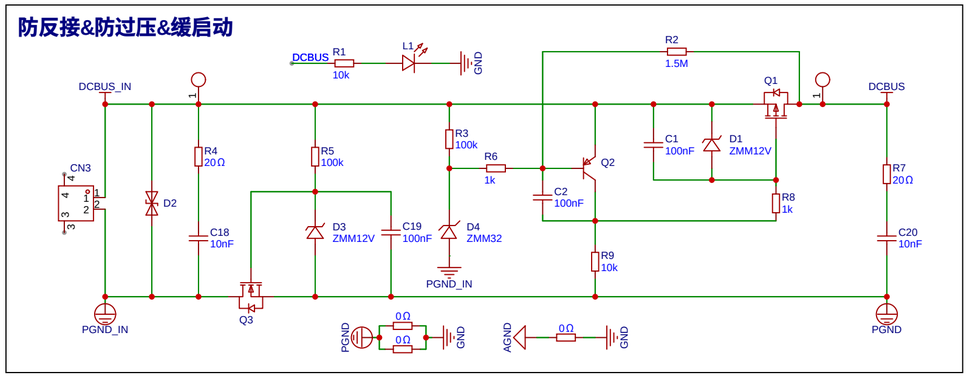
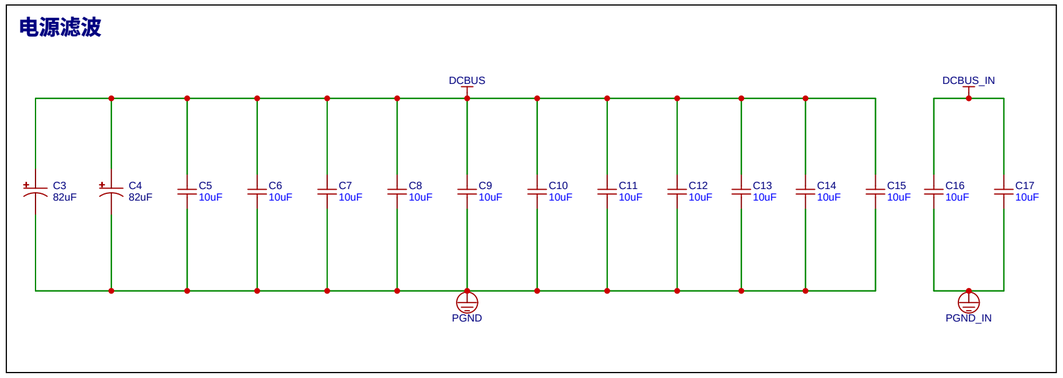
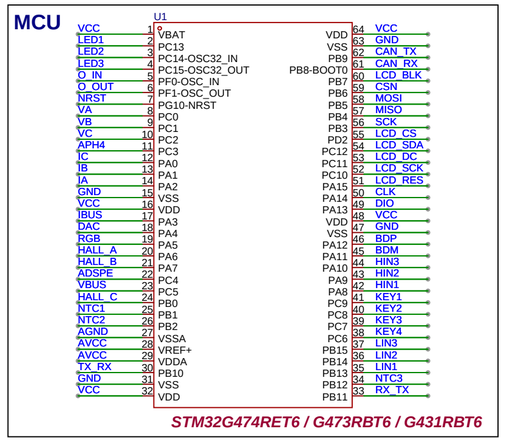
防反接&防过压&缓启动



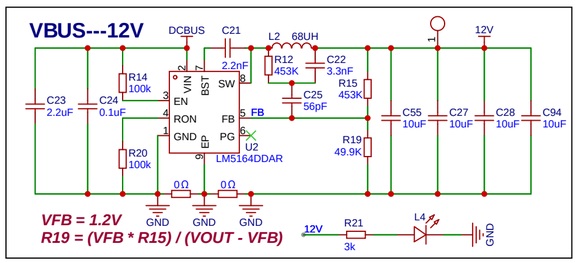
电源滤波



MCU

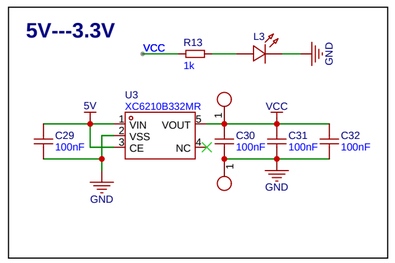


VBUS---12V

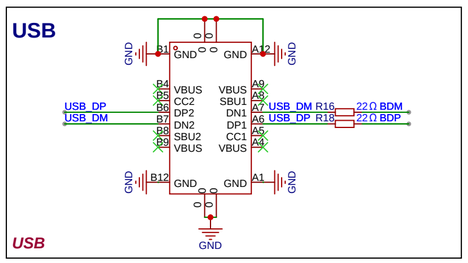


VFB = 1.2V
R19 = (VFB * R15) / (VOUT - VFB)

5V---3.3V

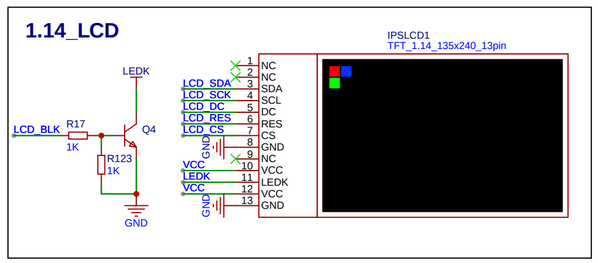


USB

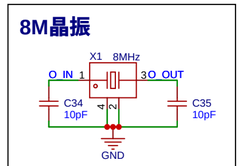


USB

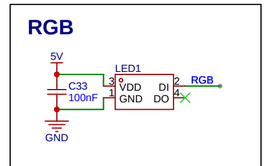
1.14_LCD



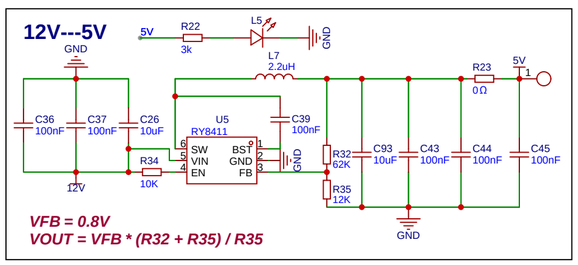
8M晶振



RGB

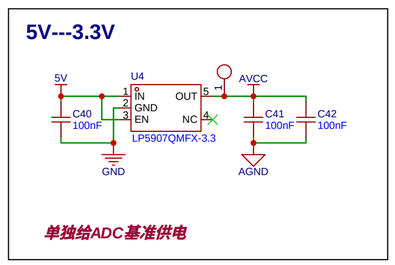


12V---5V



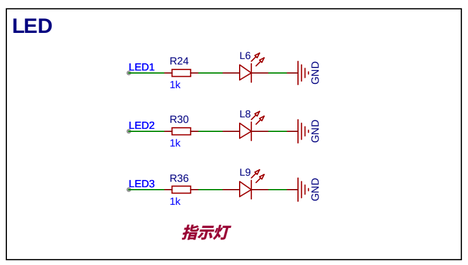
VFB = 0.8V
VOUT = VFB * (R32 + R35) / R35

5V---3.3V



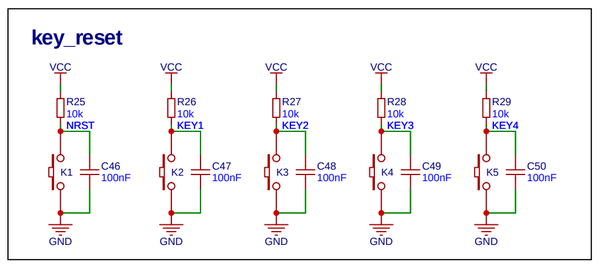
单独给ADC基准供电

LED

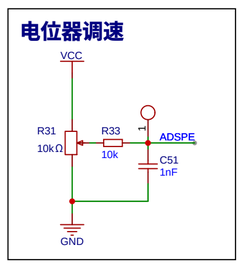


指示灯

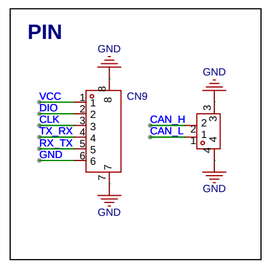
key_reset



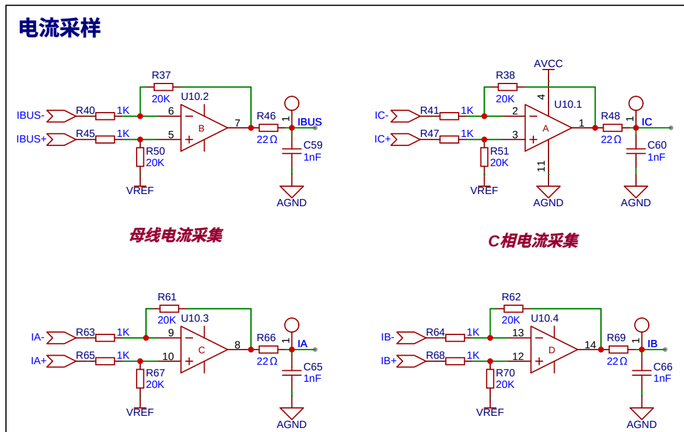
电位器调速



PIN

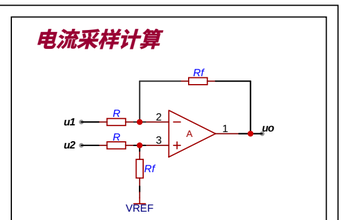


电流采样



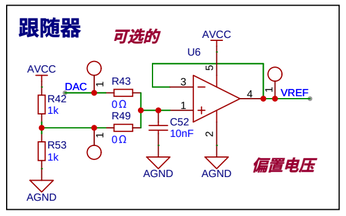
母线电流采集　C相电流采集　A相电流采集　B相电流采集

电流采样计算

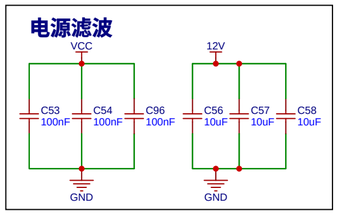


运放增益:k = Rf / R
偏置电压:VREF = 1.65V = 2048
采样电阻:Rm = 1mΩ
运放输出:uo = k * (u2 - u1)
u2 - u1 = I * Rm
uo = (adc_val - 2048) * 3.3V/4096
采集电流:I = uo / (Rm * k)

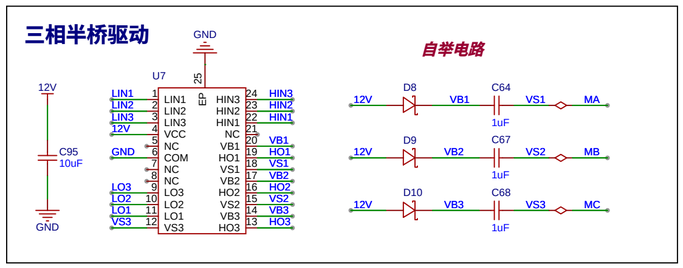
跟随器



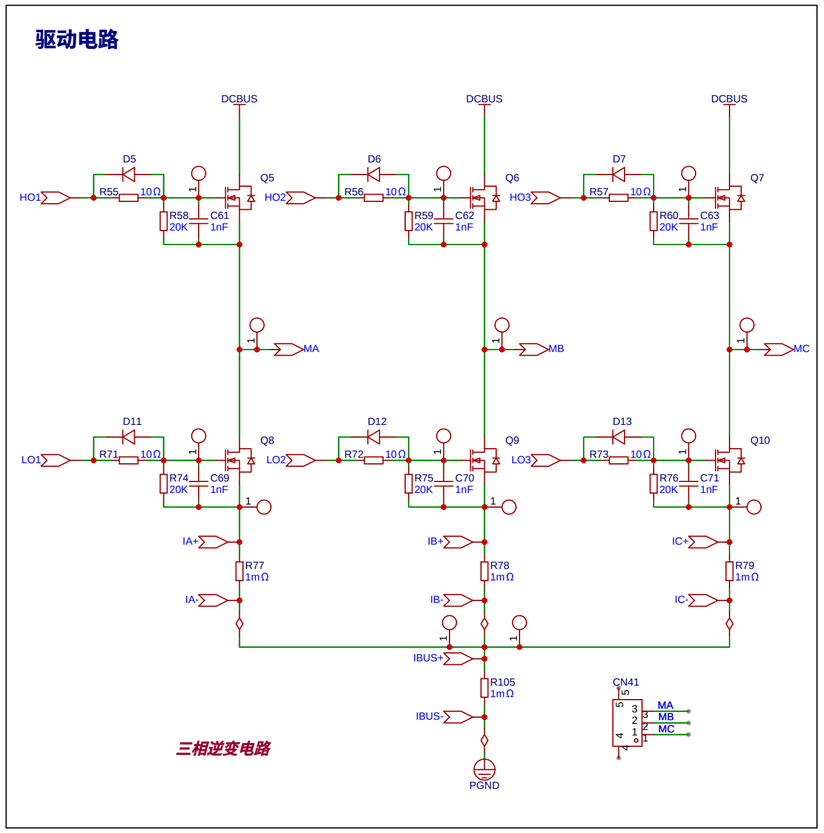
电源滤波



三相半桥驱动

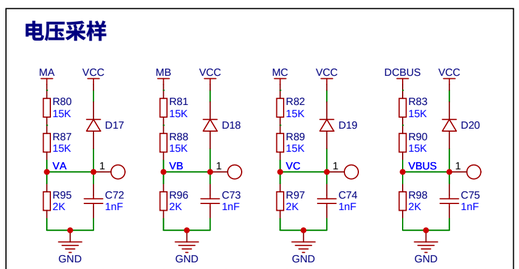


驱动电路



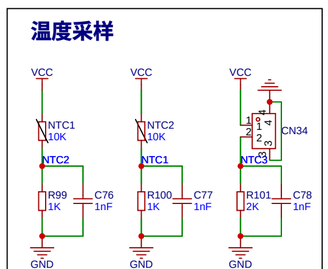
三相逆变电路

电压采样

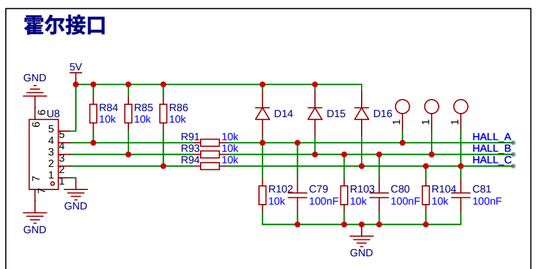


A相电压采样　B相电压采样　C相电压采样　母线电压采样

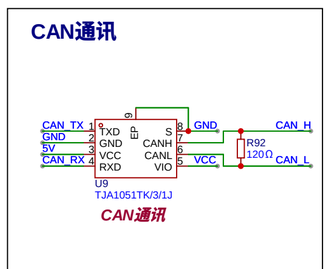
温度采样



霍尔接口

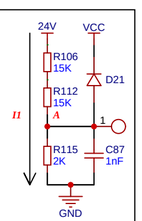


CAN通讯

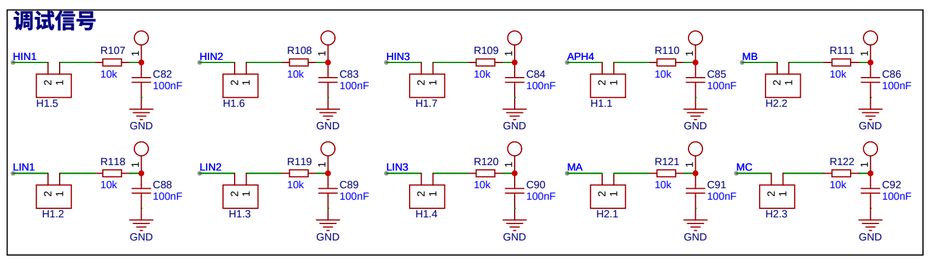


CAN通讯

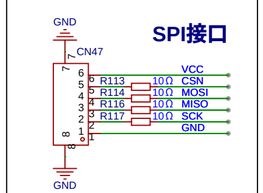
采样点最高电压为: 3.3V + 二极管管压降
采样点电压应接近3.3V的一半,分辨率高，易采样
I1电流: 24V/(15k+15k+2k) = 0.75mA
A点电压: I1 * 2k = 1.5V
缩小电压系数: k = (15k+15k) / 2k
VA = adc_val * (3.3V/4096) * k



调试信号



SPI接口



Schematic	Schematic2_2	Update Date	2023-10-27
		Create Date	2023-10-26
Page	P1	Part Number	
Drawed	foc_driver		
Reviewed			
	VER	SIZE	PAGE 1 OF 1
嘉立创EDA			嘉立创EDA